Melbourne Café prediction

By Jean Adrien Auguste

Table of Contents

Use case: ***2***

Data from the MOP: **2**

User Story: ***2***

Exploratory data analysis (EDA): ***2***

Steps in building the projects: ***3***

Step 1: Accessing the datasets. 3

Step 2: Data wrangling. 4

Step 3: Merging datasets: 4

Step 4. Understanding the correlation between the features and the target. 5

Step 5. Machine learning selection. 6

Use case:

The project is aimed at predicting the number café in Melbourne inner suburb based on three factors residential dwelling dataset, industries and number of café present in the suburb.

Data from the MOP:

Residential Dwelling:

https://data.melbourne.vic.gov.au/explore/dataset/residential-dwellings/information/

Café, restaurant, bistro seats:

https://data.melbourne.vic.gov.au/explore/dataset/cafes-and-restaurants-with-seating-capacity/information/

Industries:

https://data.melbourne.vic.gov.au/explore/dataset/employment-by-block-by-clue-industry/table/

User Story:

As an investor looking into opening a new café, I would like to know what are the industries, and numbers of residential dwelling within the suburb which can bring customers to my café. Likewise, I would also like to know the number of cafés already present in the suburb as they present competition.

The product needs to be a prediction of the last 3 to 4 years so the investor can gauge the accuracy of the prediction.

Exploratory data analysis (EDA):

The EDA is firstly to ensure the three datasets are of the same format so they can be merged. The second step will be dealing with missing values for future machine learning application. During the EDA, the dataset needs to be checked for potential new features or feature engineering.

During the data analysis phase, correlation between the features needs to be studied to understand the impact of industries and dwelling on the café numbers.

Steps in building the projects:

Step 1: Accessing the datasets.

The first step is to apply an API call to obtain the dataset. The key steps are to obtain an API key from the MOP website and to break down the MOP dataset link into sections as shown below.

API Function for Dwelling dataset and business dataset

```
[2]: def fetch_data(base_url, dataset, api_key, num_records, max_offset,where, offset):
    all_records =[]
    #Maximum number of requests

    while True:
        # Maximum limit check
        if offset > max_offset:
            break

        # Create API request URL

        filters = f'{dataset}/records?&limit={num_records}&offset={offset}&where={where}'
        url = f'{base_url}{filters}&api_key={api_key}'

        try:
            result = requests.get(url, timeout=10)
            result.raise_for_status()
            records = result.json().get('results')
        except requests.exceptions.RequestException as e:
            raise Exception(f'API request failed: {e}')

        if records is None:
            break

        all_records.extend(records)
        if len(records) < num_records:
            break

        # next cycle offset
        offset += num_records

    # Dataframe all data
    df = pd.DataFrame(all_records)
    return df
```

```
API_KEY = 'PLEASE USE YOUR PROVIDED KEY'
BASE_URL = "https://data.melbourne.vic.gov.au/api/explore/v2.1/catalog/datasets/"
DATASET_DEVELOPMENT = 'development-activity-monitor'
DATASET_BUS = 'business-establishments-per-block-by-anzsic'
NUM_RECORD = 100
MAX_OFFSET_DEV = 1420
MAX_OFFSET_BUS = 12394
OFFSET = 0
# General select and group_by for Business and Dwelling
WHERE_DEV = ''
WHERE_BUS_2011 = 'census_year%20%3E%202010'
WHERE_BUS_2010 ='census_year%20%3C%202011'
```

Step 2: Data wrangling.

During the data wrangling phase, it is important to check each dataset individually to ensure there are no null values. For instance, we checked over the feature status in the dwelling dataset to only keep the completed dwelling data. After which a handful of features were selected from the dataset and a new feature was created as Total.

[15]:
```
#2. Identify required fields
df_dev.iloc[:,[3,4,13,14,15,16,17,18]].head(10)
```

[15]:

	year_completed	clue_small_area	resi_dwellings	studio_dwe	one_bdrm_dwe	two_bdrm_dwe	three_bdrm_dwe	student_apartments
0	2006	North Melbourne	17	0	0	0	0	0
1	2005	North Melbourne	12	0	0	0	0	0
2	2013	West Melbourne (Residential)	3	0	0	0	3	0
3	2014	West Melbourne (Residential)	28	0	17	11	0	0
4	2007	North Melbourne	0	0	0	0	0	0
5	2002	Carlton	14	0	0	0	0	0
6	2015	Melbourne (Remainder)	0	0	0	0	0	0
7	2009	West Melbourne (Residential)	0	0	0	0	0	0
8	2010	Melbourne (CBD)	0	0	0	0	0	0
9	2006	Carlton	0	0	0	0	0	70

[16]:
```
#3. Produce the total Number of residential dwelling for individual clue_small_ares for each year.
df_dev['Total'] = df_dev.iloc[:,[13,14,15,16,17,18]].sum(axis=1)
```

Step 3: Merging datasets:

Before merging datasets, specific features need to be identified to merge on. Below, we developed the merge on the suburb name(clue_small_area) and the year(cenus_year).

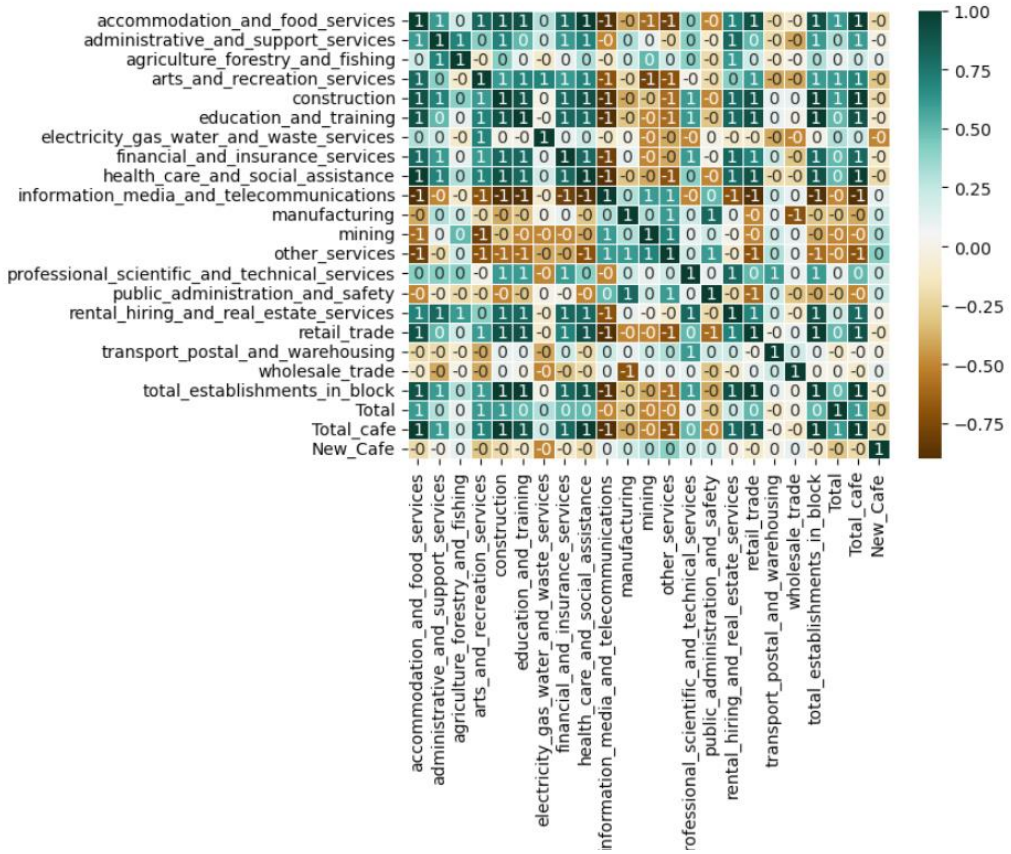
[34]:
```
#2. Merge data on year and clue_small_area
data_merged = pd.merge(df_bus, df_dev,on=['census_year','clue_small_area'],how='inner')
data_merged.shape
```

[34]:
```
(210, 23)
```

Step 4. Understanding the correlation between the features and the target.

During this phase, it is important to identify the features required to perform the future prediction. We employed a function which returns a visual of the correlation matrix and a new dataset based on the target correlation value to the feature where feature with a selected minimum correlation is selected.

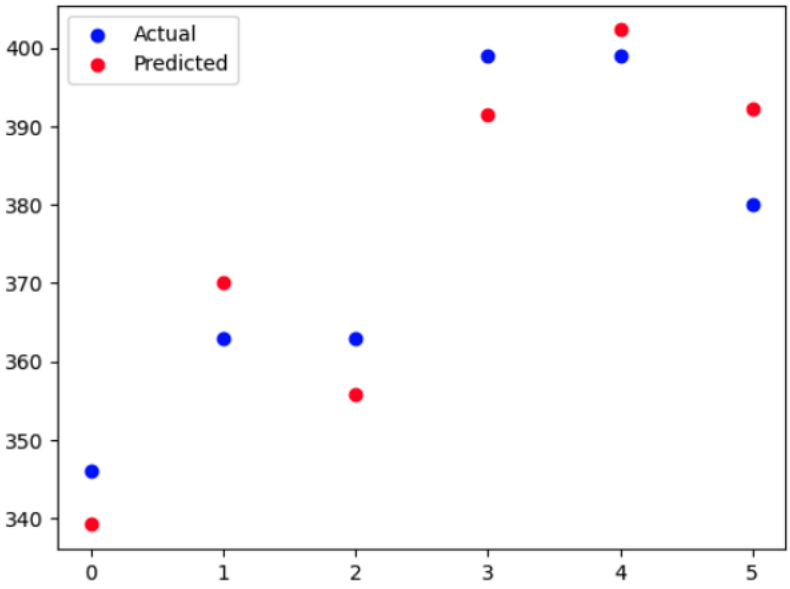
```
246]: Southbank =heat_map(data,'Southbank')
      Southbank.shape
```



Step 5. Machine learning selection.

This phase was the most time consuming as various machine learning and deep learning algorithms were experimented with to obtain an accurate reading. Machine learning was the best fit for the project as the dataset was too small for deep learning. In this phase, we applied the machine learning algorithm along with parameter tunning to improve the prediction.

The best prediction obtained was from ElacticNet a regression model with 84% R-squared value.

```
[314]: model_elastic_best(10.0,0.99, X_train, X_test, Y_train, Y_test, pred, name)
```

```
Mean Suqared Error: 61.13695634511248
R-squared 0.8409272601601583
The predicted number of new cafe in Carlton is [400]
```